

Proof of Smoothness and Uniqueness for Morphic-Regularized Navier-Stokes Equations

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Introduction

In this final step, we address the uniqueness of solutions to the morphic-regularized Navier-Stokes equations. While the introduction of the regularization term $\epsilon \mathbf{R}(\mathbf{u})$ provides smooth solutions, we now focus on proving that these solutions are unique, ensuring that the system yields a single, well-defined solution for any given set of initial conditions.

Step 5: Uniqueness of Solutions

The uniqueness of the Navier-Stokes solutions is a central challenge. To prove uniqueness for the morphic-regularized system, we must show that if two solutions exist, their difference diminishes over time, ultimately converging to zero. This step involves comparing two potential solutions $\mathbf{u}_1$ and $\mathbf{u}_2$, and demonstrating that the difference between them vanishes.

Consider Two Solutions

Let $\mathbf{u}_1(x, t)$ and $\mathbf{u}_2(x, t)$ be two smooth solutions to the morphic-regularized Navier-Stokes equations. We define their difference as:

$$\mathbf{w}(x, t) = \mathbf{u}_1(x, t) - \mathbf{u}_2(x, t).$$

Taking the difference between the two versions of the Navier-Stokes equations gives:

$$\frac{\partial \mathbf{w}}{\partial t} + (\mathbf{u}_1 \cdot \nabla) \mathbf{w} - (\mathbf{u}_2 \cdot \nabla) \mathbf{w} = \nu \nabla^2 \mathbf{w} + \epsilon \mathbf{R}(\mathbf{w}).$$

Since $\mathbf{u}_1$ and $\mathbf{u}_2$ are both incompressible, we also have:

$$\nabla \cdot \mathbf{w} = 0.$$

Energy Estimate for $\mathbf{w}(x, t)$

We now derive an energy estimate for $\mathbf{w}(x, t)$. Taking the L^2 -norm of the difference $\mathbf{w}$ yields the energy:

$$E_w(t) = \frac{1}{2} \int_{\mathbb{R}^n} |\mathbf{w}(x, t)|^2 dx.$$

Taking the time derivative of $E_w(t)$ gives:

$$\frac{d}{dt} E_w(t) = \int_{\mathbb{R}^n} \mathbf{w}(x, t) \cdot \frac{\partial \mathbf{w}}{\partial t} dx.$$

Substituting the equation for $\frac{\partial \mathbf{w}}{\partial t}$, we get:

$$\frac{d}{dt} E_w(t) = \int_{\mathbb{R}^n} \mathbf{w} \cdot (-(\mathbf{u}_1 \cdot \nabla) \mathbf{w} + (\mathbf{u}_2 \cdot \nabla) \mathbf{w} + \nu \nabla^2 \mathbf{w} + \epsilon \mathbf{R}(\mathbf{w})) dx.$$

The convective terms cancel out due to incompressibility, and the remaining terms yield:

$$\frac{d}{dt} E_w(t) = -\nu \|\nabla \mathbf{w}\|_{L^2}^2 + \epsilon \int_{\mathbb{R}^n} \mathbf{w} \cdot \mathbf{R}(\mathbf{w}) dx.$$

Dissipation of Energy Difference

The viscous term $-\nu \|\nabla \mathbf{w}\|_{L^2}^2$ ensures dissipation of the energy difference over time. The regularization term $\epsilon \mathbf{R}(\mathbf{w})$ provides additional smoothing, dissipating any remaining energy in $\mathbf{w}$. Since the regularization term is dissipative, we have:

$$\int_{\mathbb{R}^n} \mathbf{w} \cdot \mathbf{R}(\mathbf{w}) dx \leq 0,$$

which guarantees further dissipation.

Thus, the total dissipation is given by:

$$\frac{d}{dt} E_w(t) \leq -\nu \|\nabla \mathbf{w}\|_{L^2}^2.$$

This implies that the energy difference $E_w(t)$ decreases over time, driving $\mathbf{w}(x, t)$ toward zero.

Uniqueness of Solutions

Since $E_w(t) \geq 0$ and $\frac{d}{dt} E_w(t) \leq 0$, it follows that $E_w(t) \rightarrow 0$ as $t \rightarrow \infty$. Therefore, the difference between the two solutions $\mathbf{u}_1$ and $\mathbf{u}_2$ vanishes over time, meaning that:

$$\mathbf{u}_1(x, t) = \mathbf{u}_2(x, t) \quad \text{for all } t.$$

This establishes that the morphic-regularized Navier-Stokes equations have a unique solution for all initial conditions.

1 Conclusion of Step 5

We have demonstrated that the morphic-regularized Navier-Stokes equations admit a unique smooth solution for all time. The dissipation introduced by the regularization term ensures that any two solutions will converge to the same result, proving uniqueness.